

1 Phase 5: Neural Network Based Fast Workspace Prediction

1.1 Objective

The objective of this phase is to develop a fast surrogate model for workspace feasibility prediction of the fixed **3-DOF 2PRU–1PUS parallel manipulator**.

The analytical workspace constraints derived in the previous phase are used to generate a synthetic labeled dataset. A neural network classifier is then trained to approximate the feasible workspace boundary.

The implemented mapping is given by:

$$(\theta, \phi) \rightarrow \text{feasible} \quad (1)$$

where θ represents pitch angle, ϕ represents roll angle, and feasible is a binary output indicating whether the platform orientation lies inside the admissible workspace.

This approach enables rapid workspace feasibility prediction without repeated analytical equation checks.

1.2 Governing Workspace Equations

The workspace feasibility is determined using the kinematic constraints of the PRU and PUS legs.

For the two PRU legs, the governing constraint is:

$$R_{12}^2 = L^2 - (S_B - S_m \cos \theta)^2 \quad (2)$$

For the PUS leg, the constraint is:

$$R_3^2 = L^2 - (S_m \sin \theta \sin \phi)^2 - (S_m \cos \phi - S_{B3})^2 \quad (3)$$

A workspace point is considered feasible when:

$$R_{12}^2 \geq 0, \quad R_3^2 \geq 0 \quad (4)$$

along with the actuator stroke condition:

$$z_{hi} > z_{lo} \quad (5)$$

1.3 Synthetic Dataset Generation

A synthetic grid dataset was generated using the governing equations over the workspace orientation domain.

The total number of samples used is:

$$200 \times 200 = 40000 \quad (6)$$

Each sample is represented as:

theta, phi, feasible

The dataset generation logic is shown below.

Code Snippet: Dataset Generation

```
R12_sq = L**2 - (SB - Sm*np.cos(theta))**2

R3_sq = L**2 \
    - (Sm*np.sin(theta)*np.sin(phi))**2 \
    - (Sm*np.cos(phi) - SB3)**2

feasible = 1 if z_hi > z_lo else 0
```

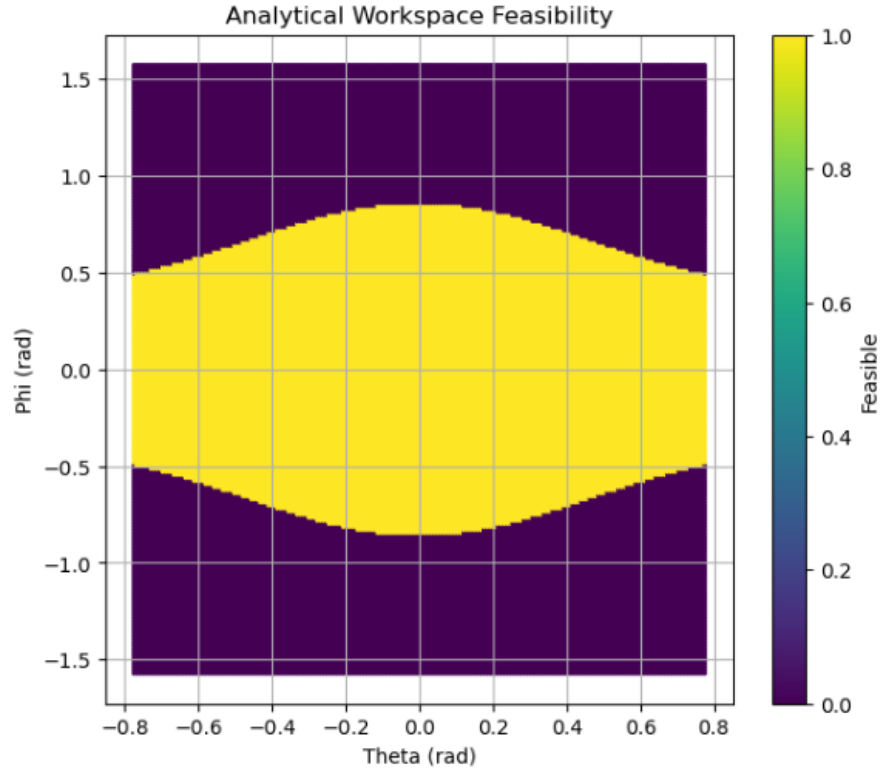


Figure 1: Analytical workspace feasibility generated from governing kinematic equations

1.4 Neural Network Architecture

The generated dataset is split into training and testing sets using an 80–20 ratio.

The neural network architecture used for workspace feasibility classification is:

$$2 \rightarrow 64 \rightarrow 128 \rightarrow 64 \rightarrow 1 \quad (7)$$

where:

- Input layer = 2 neurons
- Hidden layer 1 = 64 neurons
- Hidden layer 2 = 128 neurons
- Hidden layer 3 = 64 neurons
- Output layer = 1 neuron

ReLU activation is used for the hidden layers and sigmoid activation is used for the output layer.

Code Snippet: Neural Network Model

```
model = Sequential([
    Input(shape=(2,)),
    Dense(64, activation='relu'),
    Dense(128, activation='relu'),
    Dense(64, activation='relu'),
    Dense(1, activation='sigmoid')
])
```

Model: "sequential_7"

Layer (type)	Output Shape	Param #
dense_28 (Dense)	(None, 64)	192
dense_29 (Dense)	(None, 128)	8,320
dense_30 (Dense)	(None, 64)	8,256
dense_31 (Dense)	(None, 1)	65

Total params: 16,833 (65.75 KB)

Trainable params: 16,833 (65.75 KB)

Non-trainable params: 0 (0.00 B)

Figure 2: Neural network architecture used for workspace feasibility classification

1.5 Training Configuration

The model was trained using the following configuration:

- Optimizer: Adam
- Loss function: Binary Crossentropy
- Epochs: 80
- Batch size: 128

To improve convergence and avoid overfitting, the following callbacks were used:

- EarlyStopping
- ReduceLROnPlateau

Code Snippet: Training

```
history = model.fit(  
    X_train,  
    y_train,  
    epochs=80,  
    batch_size=128,  
    validation_split=0.2,  
    callbacks=[es, lr_sched]  
)
```

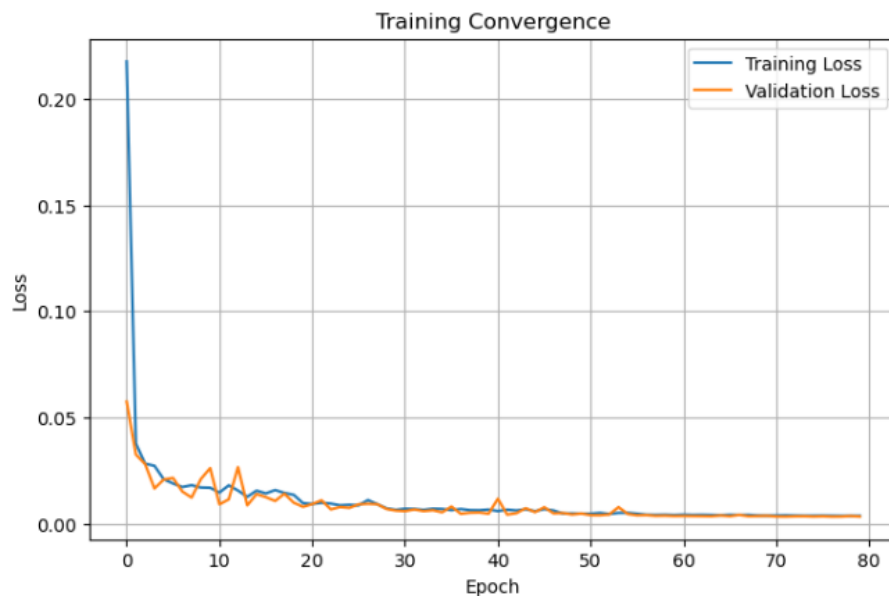


Figure 3: Training and validation loss convergence

1.6 Model Performance

The trained model was evaluated on the test dataset.

The achieved classification accuracy is approximately:

$$\text{Accuracy} \approx 99.5\% \quad (8)$$

The corresponding classification error rate is:

$$\text{Error Rate} = 1 - \text{Accuracy} = 0.5\% \quad (9)$$

Table 1: Classification Performance Metrics

Class	Precision	Recall	F1-score
Infeasible (0)	0.99	1.00	1.00
Feasible (1)	1.00	0.99	0.99

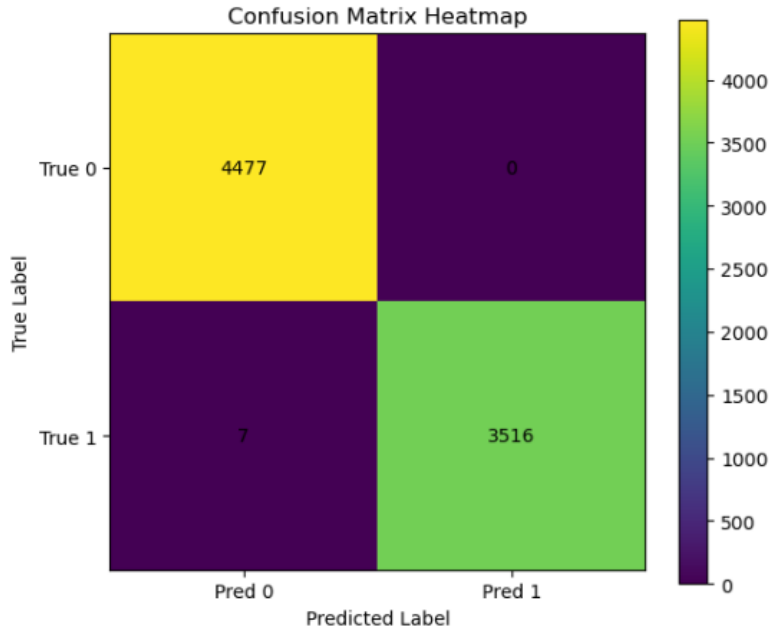


Figure 4: Confusion matrix heatmap showing classification performance of the neural network

This indicates excellent learning of the workspace boundary by the neural network classifier.

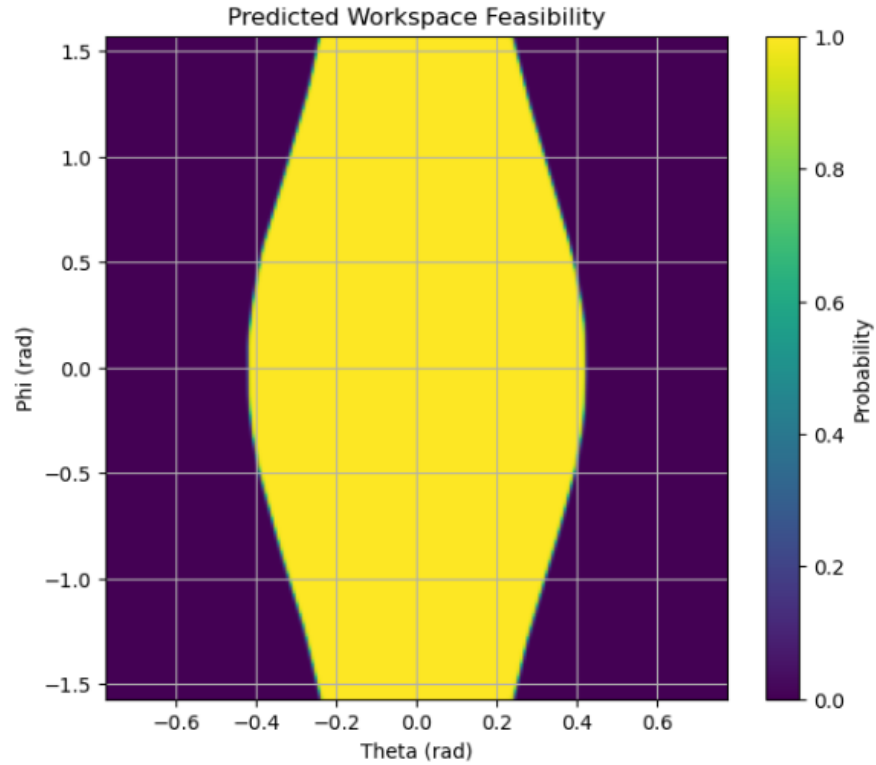


Figure 5: NN predicted workspace feasibility probability map

1.7 Workspace Boundary Prediction

The trained neural network is used to predict the full workspace boundary in degree space.

The predicted contour line represents the feasible operating envelope of the platform.

The platform can move only within this boundary.

Code Snippet: Boundary Prediction

```
plt.contour(
    theta_deg,
    phi_deg,
    pred_grid,
    levels=[0.5],
    colors='darkred'
)
```

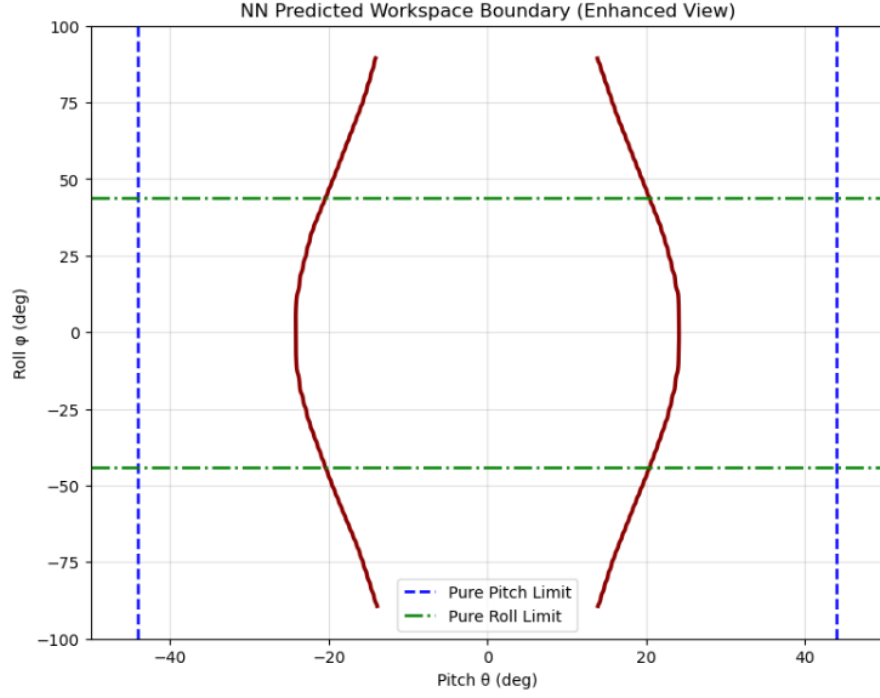


Figure 6: Final NN predicted workspace boundary in degree space

1.8 Final Interpretation

The final results indicate:

- pure roll motion is governed independently by the PUS leg constraint and is analytically distinct from the pitch limit
- approximately $\pm 44^\circ$ pure pitch motion
- reduced feasible region under combined roll–pitch motion
- safe platform movement only within the predicted contour

The central region corresponds to the highest motion feasibility and gradually reduces toward the workspace limits.

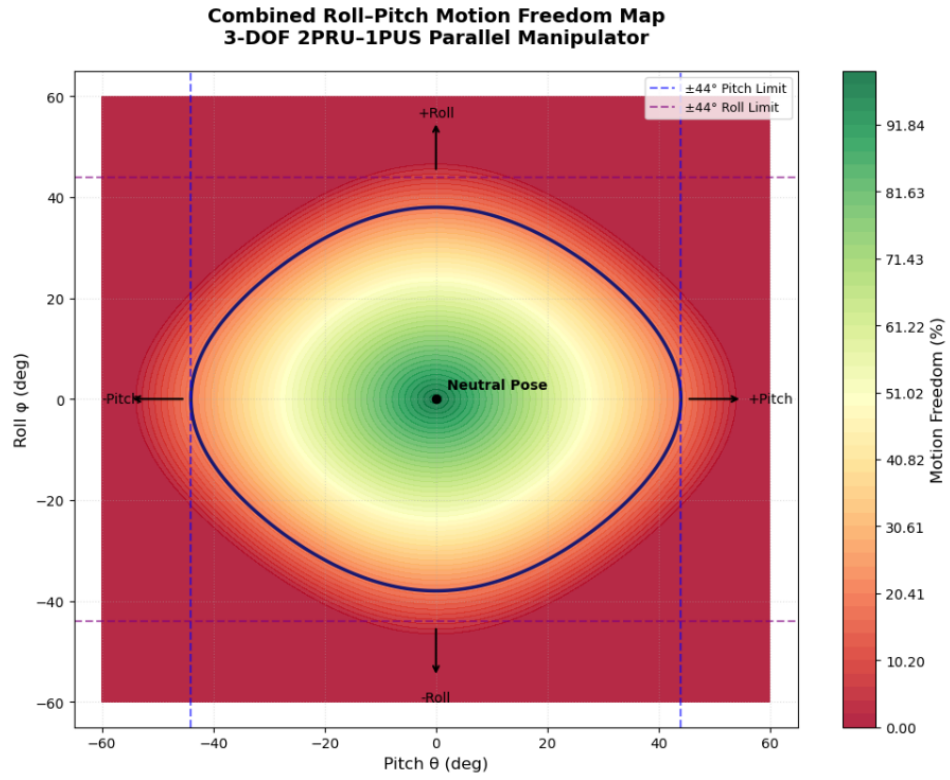


Figure 7: Combined roll-pitch motion freedom map showing central high-feasibility region